

Name: \_\_\_\_\_

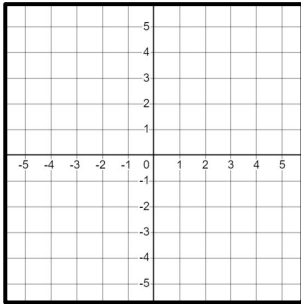


# The equation of a straight line

## Task A: Different forms of linear relationships

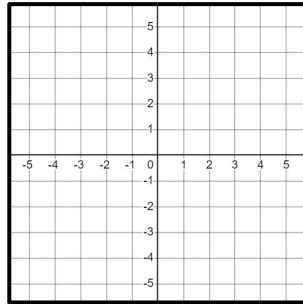
1) Plot each equation and identify the gradient and y-intercept.

a)  $x + y = 5$



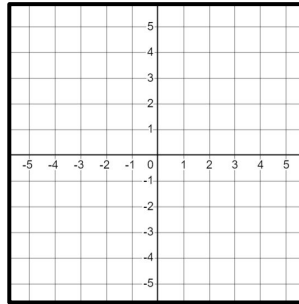
gradient =  
y-intercept =

b)  $y = 2x - 5$



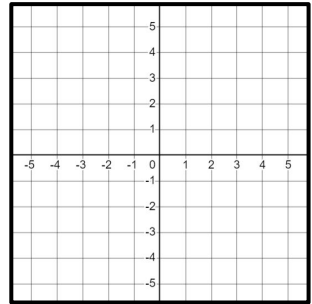
gradient =  
y-intercept =

c)  $y = 5 - 2x$



gradient =  
y-intercept =

d)  $2x + 2y = 10$

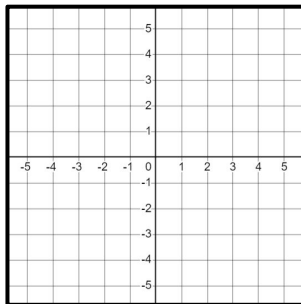


gradient =  
y-intercept =

2) Match up any equations that represent the same relationship

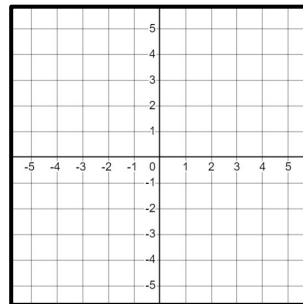
3) Plot each equation and identify the gradient and y-intercept.

a)  $x + y = 4$



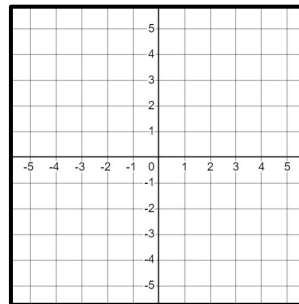
gradient =  
y-intercept =

b)  $y = 4 - x$



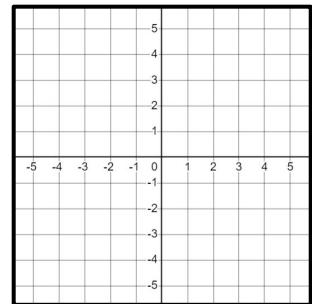
gradient =  
y-intercept =

c)  $y = \frac{1}{2}x + 2$



gradient =  
y-intercept =

d)  $2y + x = 4$



gradient =  
y-intercept =

4) Match up any equations that represent the same relationship



**Task B: Gradient and y-intercept from  $y = mx + c$** 

1) Fill in the table with the gradient and y-intercept for each equation.

Write the y-intercept as a coordinate.

| equation              | gradient | y-intercept |
|-----------------------|----------|-------------|
| a) $y = 8x + 4$       |          |             |
| b) $y = 3x - 7$       |          |             |
| c) $y = -5x + 1$      |          |             |
| d) $y = 10 - 3x$      |          |             |
| e) $y = x$            |          |             |
| f) $y = -x$           |          |             |
| g) $y = 5$            |          |             |
| h) $y = p + (q + 3)x$ |          |             |

2) Fill in the missing pieces of information for each question.

| equation                                                      | gradient                   | y-intercept                                                 |
|---------------------------------------------------------------|----------------------------|-------------------------------------------------------------|
| a) $y = \underline{\hspace{1cm}}x + 4$                        | 9                          | (0, $\underline{\hspace{1cm}}$ )                            |
| b) $y = \frac{1}{2}x - 6$                                     | $\underline{\hspace{1cm}}$ | ( $\underline{\hspace{1cm}}$ , $\underline{\hspace{1cm}}$ ) |
| c) $y = 3(4x + 2)$                                            | $\underline{\hspace{1cm}}$ | ( $\underline{\hspace{1cm}}$ , $\underline{\hspace{1cm}}$ ) |
| d) $y = 10 \underline{\hspace{1cm}}$                          | -1                         | (0, $\underline{\hspace{1cm}}$ )                            |
| e) $5(1 - 2x) = y$                                            | $\underline{\hspace{1cm}}$ | ( $\underline{\hspace{1cm}}$ , $\underline{\hspace{1cm}}$ ) |
| f) $8 - \underline{\hspace{1cm}}x = y$                        | -5                         | (0, $\underline{\hspace{1cm}}$ )                            |
| g) $y = \underline{\hspace{1cm}} + \underline{\hspace{1cm}}x$ | $t$                        | (0, $u$ )                                                   |